

NEUROMUSCULAR DISORDERS

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Disorders with :

- Neurological component affecting the : Spinal cord & peripheral nerves.
- muscular component : muscle weakness.

Spinal Muscular Atrophy (SMA)

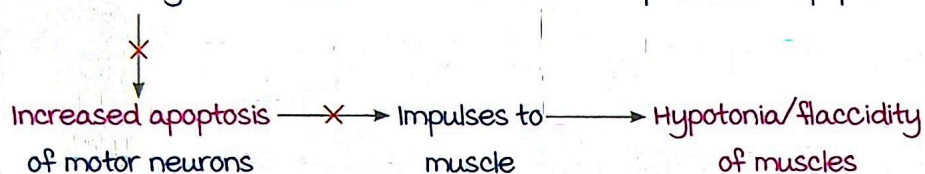
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Pathogenesis :

- Autosomal recessive disorder : Chromosome 5q → **SMN gene** (survival motor neuron) → SMN protein.

SMN gene :

- Role of SMN gene : Protects motor neurons from premature apoptosis.



Types of SMN gene :

- **SMN 1 gene** (main gene) : Codes for fully functional protein, defective in SMA.
- **SMN 2 gene** : Lacks exon 7 → Codes for dysfunctional protein.

Clinical features :

- As seen in Type I SMA (Spinal muscular atrophy).

1. Hypotonia :

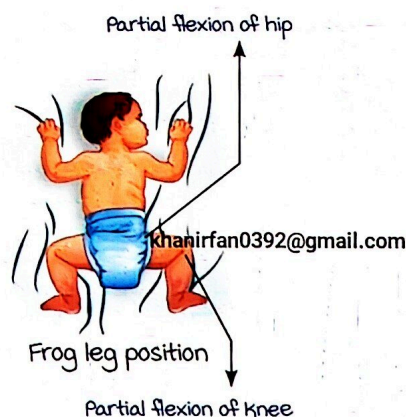
- **Frog leg posture.**
 - ↓/absent spontaneous limb movements.
 - No resistance to passive flexion.
 - **Ragged doll appearance.**
- } D/t ↓↓ hypotonia

2. LMN type of weakness :

- Deep tendon reflexes (DTR) ↓↓.
- Disuse atrophy of muscle.

3. Other features :

- Extraocular muscles (EOM) not involved.
- Tongue fasciculations (+).
- mentation is normal.



Note :

EOM involvement : myasthenia gravis.

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Complications :

Hypotonia of muscles involved in swallowing → Aspiration of feeds → **Aspiration pneumonitis** :
m/c cause of morbidity/ mortality in SMA

Clinical types of SMA (based on severity) :

Decreasing severity ↓

Type	Age of onset	motor milestones	Lifespan
0 (Fatal)	Antenatal (↓ in fetal movements)	-	Few weeks
1 (Werdnig Hoffman disease) • Classical • m/c	< 6 months	upto head control	< 2 years
2 (Dubowitz disease)	< 18 months	upto : Sitting independently (Cannot stand/ walk)	2 nd - 3 rd decade
3 (Kugelberg - Welander disease)	> 18 months	Stand, walk (Partially ambulant)	Normal
4	Adolescence/adult	Fully ambulant	Normal

management :

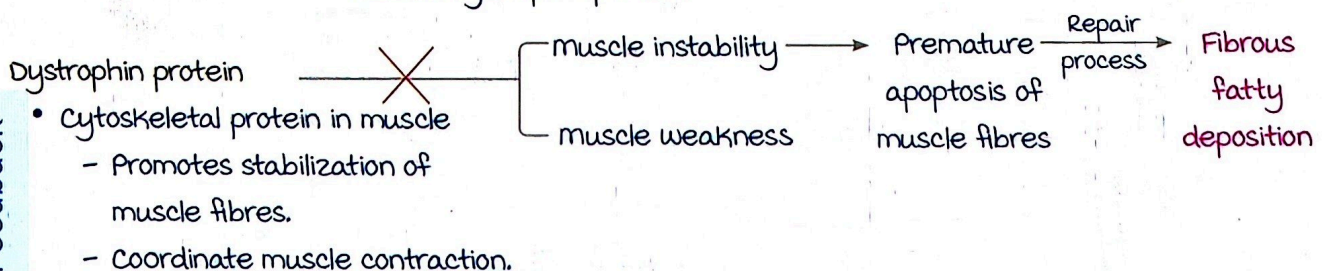
- Not a curable condition.
- Recent advances :

Drug	Therapy	mechanism of action
Zolgensma	Gene therapy : SMN gene incorporated with vector (Adenovirus)	• SMN gene incorporates with a RNA vector.
Nusinersen (Antisense oligonucleotide : AON)	Exon skipping therapy khanirfan0392@gmail.com	• Binds to defective exon 7 area ↓ Production of partially functioning protein
Risdiplam	Provides exon 7	• Includes exon 7 into defective area of gene.

Duchenne Muscular Dystrophy (DMD)

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- m/c variety of NMD.
mode of inheritance : X-linked recessive (xp_{al}).
mutated gene : **Dystrophin gene affected** (m/c : nonsense mutation) → Absent dystrophin protein.

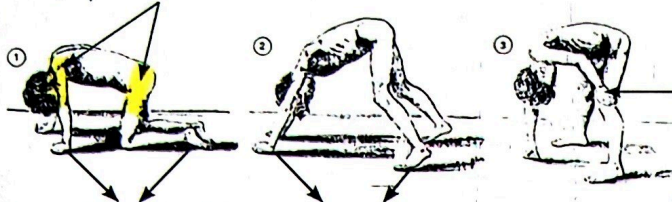


Feedback

Clinical features :

1. Early features : Delay in attaining motor milestones.
2. Borderline low IQ (Intellectual disability).
3. Proximal muscle weakness : 3-4 years of age.
 - Hip : **waddling gait** (Swaying from one side to another).
 - **Gower's sign** seen : Characteristic but **not diagnostic**.

Shoulder & hip muscle affected



Proximal muscles of lower limb supported by distal muscles of upper limb

Distal muscles (Hand, foot) not affected Supporting on distal muscles

4. **Calf muscle pseudo hypertrophy**.



Fibrofatty deposition

Plateauing of illness between 4-6 yrs

5. Progression : Dangerous manifestation

- Spine muscles : Scoliosis + lordosis.
- Bulbar muscles : Dysfunction of pharyngeal/laryngeal muscles.
- Cardiac + respiratory muscle insufficiency (**m/c cause of death**) → Death (20-25 years).

Investigations :

1. Serum creatinine Kinase : 10 times x normal value.
2. Gene testing : Diagnostic of **DMD**.
 - multiple PCR.
 - multiplex Ligand-dependent Probe Amplification (MLPA).
3. muscle biopsy.

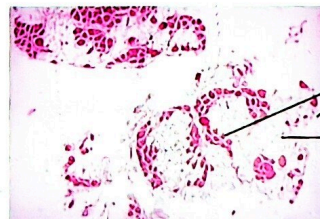
management :

- Incurable.

Steroids : ↓ rate of apoptosis.

- Deflazacort.
- Prednisolone.

Nerve treatment :

Exon skipping therapy (AON) : **Eteplisren**.**Ataluren** : Suppress nonsense mutation.

Atrophy

Fibrofatty deposition

muscle biopsy

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Becker Muscular Dystrophy

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- mild allelic (Less severe) variant of DMD (Dystrophin gene).
 - Affected gene is dystrophin gene.
 - Late onset muscle weakness (> 5 years).
 - Slow progression of symptoms → Longer lifespan : Till 40 years.
- **Early onset cardiomyopathy** (Compared to DMD).

Facioscapulohumeral Dystrophy

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- mode of inheritance : Autosomal dominant.
- Defective gene : FSHD (Facioscapulohumeral dystrophy).

Clinical features :**muscle Weakness :****1. Facial :**

- Defective eye closure.
- Rounded mouth.
- Puckering of lips.

2. Humeral :

- Weakness of arm muscles.
- Forearm muscles normal : **Popeye appearance.**

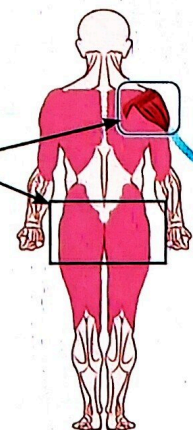
3. Scapular.**Asymmetrical winging of scapula****Limb Girdle Muscular Dystrophy**

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- mode of inheritance :
 - Autosomal recessive (m/c : Child).
 - Autosomal dominant (m/c : Adults).
- Defective proteins : Caveolin 3, calpain 3 (Cytoskeletal proteins).

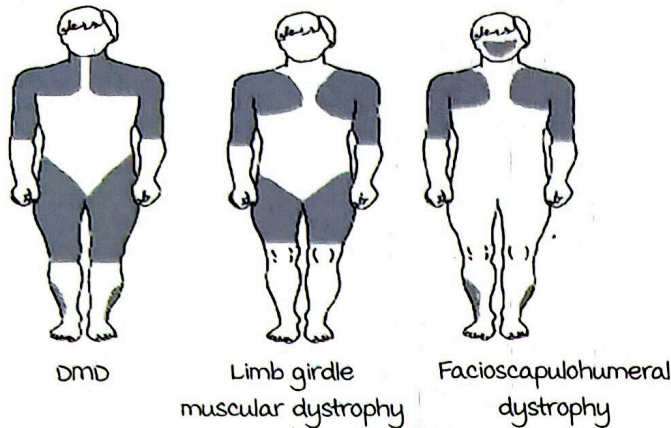
Clinical features :**Affected areas (Shoulder girdle, hip girdle)**

- Neck/bulbar muscles not affected.
- Cardiac muscle involvement (Rare).
- IQ : Normal.



Differentiating NM disorders :

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Myotonic Dystrophy Type 1

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Genetics :

- mode of inheritance : **Autosomal dominant**.
- Trinucleotide repeats : CTG > 80 (Normally < 35).
- Gene defect : **Chromosome 19** → DMPK gene.

Clinical features :

1. **HATCHET** facies :

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Late manifestations :

- Temporal hollowing (D/t wasting)
- Thinning of cheek area (D/t weakness of muscles of mastication)

Early manifestations :

1. Ptosis/ drooping of eyelids (D/t weakness of levator palpebrae superioris).
2. Inverted V appearance of upper lip.

2. **myotonia** :

- Delayed relaxation of muscles.
- Inferred by :
 - Shaking hands → Difficulty in letting hand go.
 - Percuss at base of thumb → Delayed extension after flexion.

3. Limb weakness :

- **Distal muscles affected**
 - Retarded purposeful movements with hands
 - Flattened thenar & hypothenar eminences
 - Weak foot muscles : Foot drop, abnormal gait

4. Posterior capsular cataract.

- 5. Endocrinopathies
 - Hyperinsulinism
 - Testicular atrophy
 - Adrenal gland atrophy

Feedback